Statistics

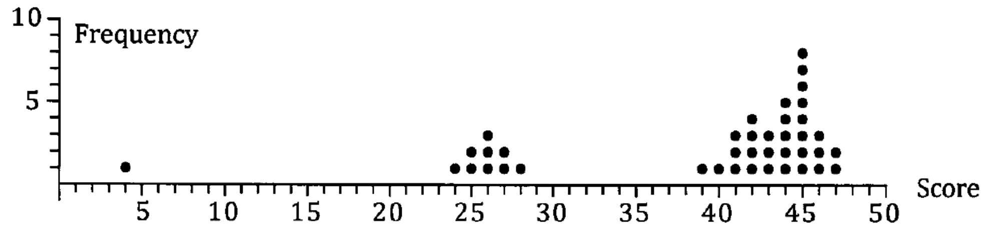
Box Plot

Different ways of representing data

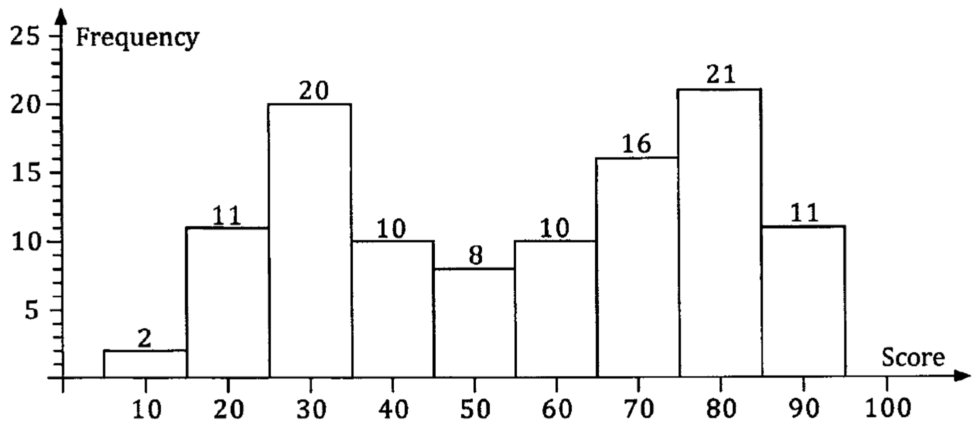
- Stem and leaf diagram

Girls		Boys
	18	6
5	17	9 3 0 5 2
8 3 5 0	16	9 6 3 2 8 5 9
7 9 5 8 5	15	9 5 9
9 8	14	9

- Dot frequency diagram



- Histogram



Box plot

- Also known as box and whisker diagram
- From the data collected, it is necessary to work out:

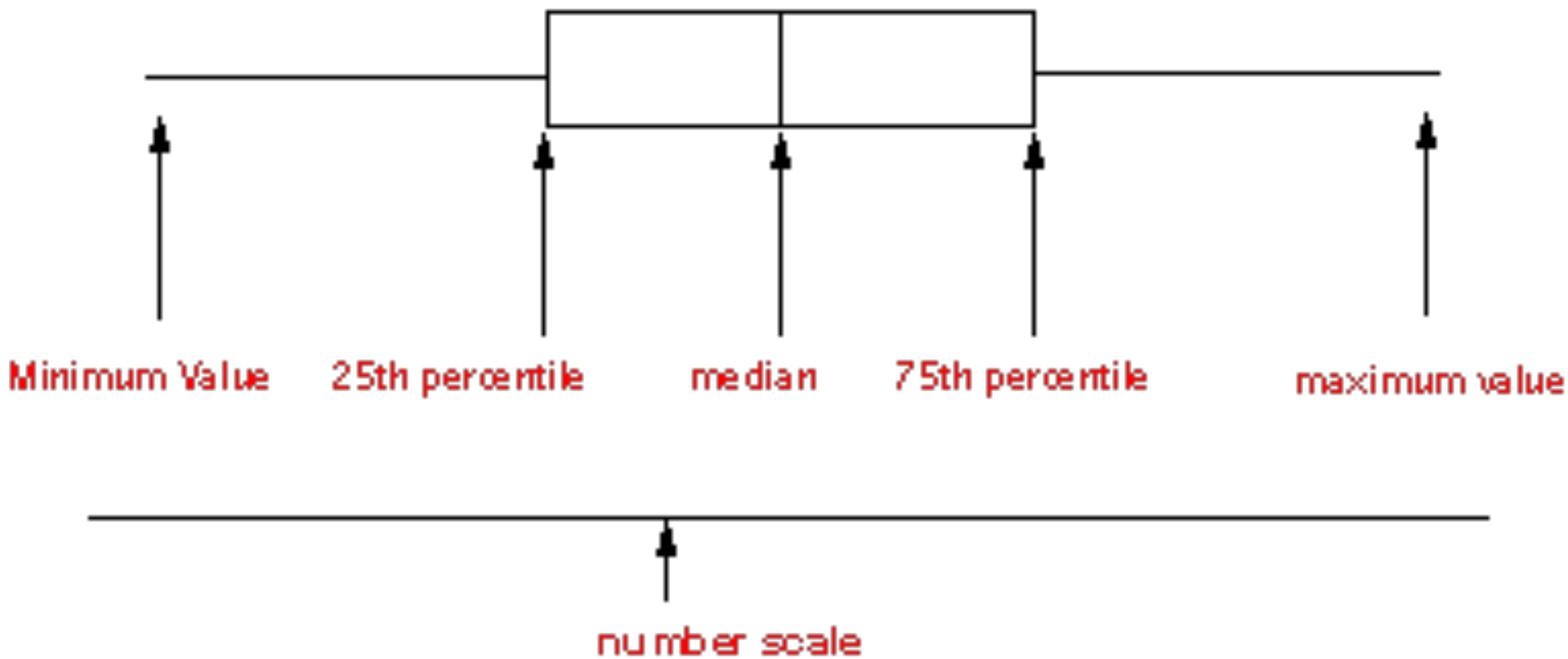
  - Minimum
  - Maximum

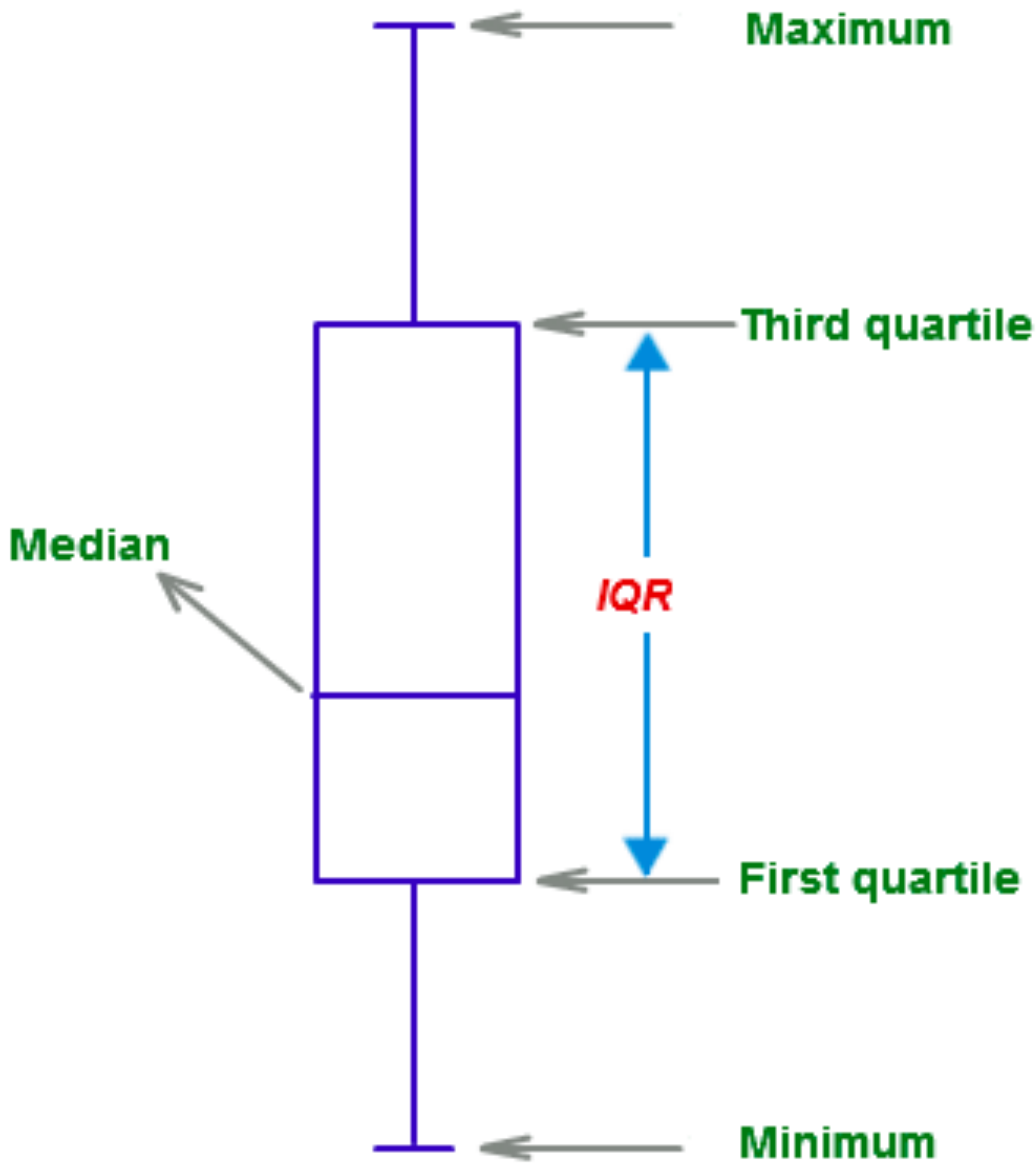
  - Median (50th percentile, second quartile)

  - Lower quartile (25th percentile, first quartile)
  - Upper quartile (75th percentile, third quartile)

  - Inter quartile range (IQR) = upper quartile – lower qurtile

Parts of a Box-n-Whisker Plot



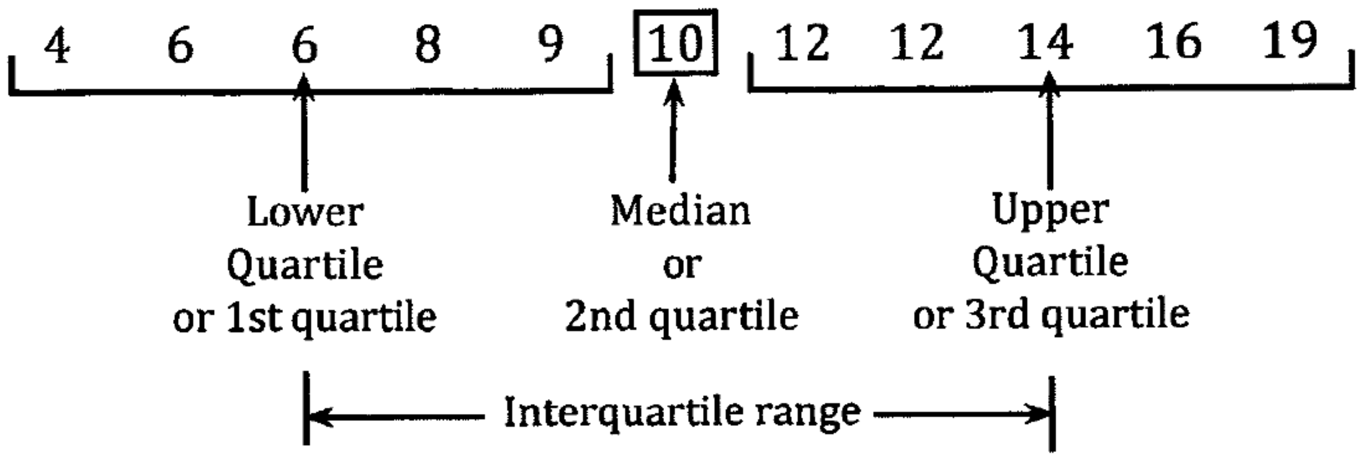

Maximum
Third quartile
Median
IQR
First quartile
Minimum

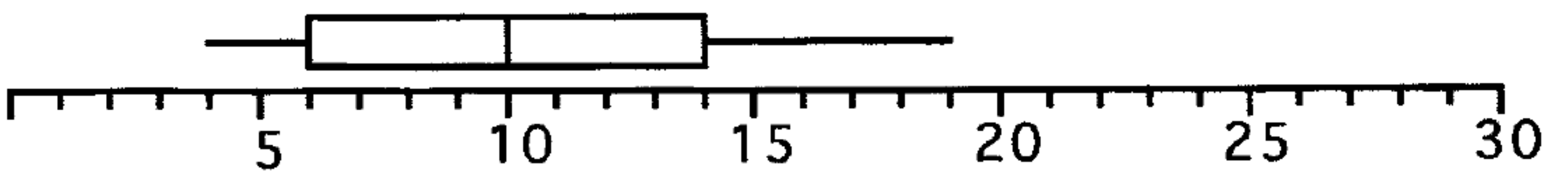
Example

For example consider the set of scores

12, 6, 10, 19, 9, 12, 4, 14, 8, 16, 6.

Listing the scores in order allows the quartiles and interquartile range to be determined:





Draw the box plot for the following data

This is the consumer rating for a particular brand of peanut butter:

71	69	60	60	57
52	34	89	69	69
67	63	57	40	

- **Step 1:** Arrange the data from smallest to largest and label them from 1 to the number of data points:

rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14
data	34	40	52	57	57	60	60	63	67	69	69	69	71	89

- **Step 2:** Find the median of the data:

rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14
data	34	40	52	57	57	60	60	63	67	69	69	69	71	89

We have 14 data values, so to compute the median of an **even number** of data values we compute the **mean of the two middle values: 7** and **8** (**14/2 = 7, 14/2 + 1 = 8**) . So with the data above we get : (60+63)/2 = **61.5**

- **Step 3:** Find the lower quartile:

data	34	40	52	57	57	60	60	63	67	69	69	69	71	89
rank	1	2	3	4	5	6	7							

The lower quartile is the median of the numbers above. Since we have an **odd** number of data points the median is the data value with rank **(7+1)/2 = 4**. So the **lower quartile is 57.**

- **Step 4:** Find the upper quartile:

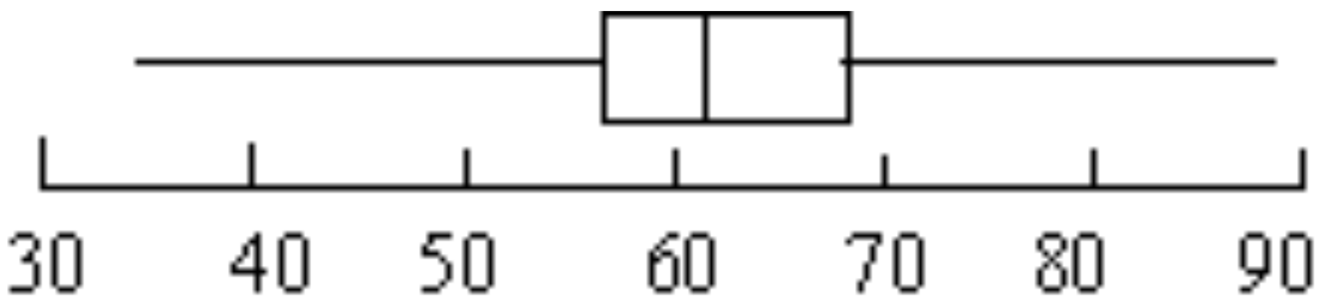
data	34	40	52	57	57	60	60	63	67	69	69	69	71	89
rank								1	2	3	4	5	6	7

The upper quartile is the median of the numbers above. Since we have an **odd** number of data points the median is the data value with rank **(7+1)/2 = 4**. So the **upper quartile is 69.**

- **Step 4:** Find the minimum and maximum value:

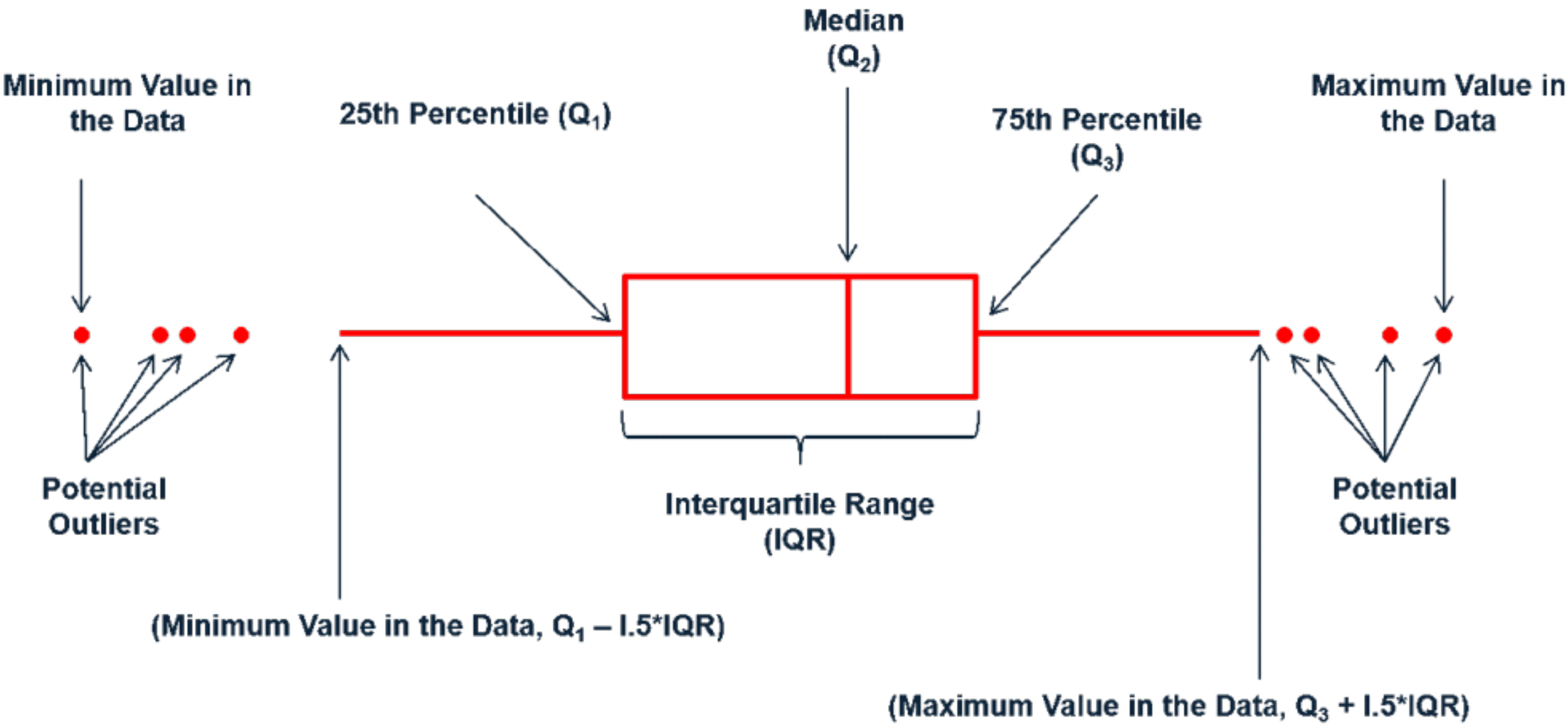
rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14
data	34	40	52	57	57	60	60	63	67	69	69	69	71	89

- **Step 5: Create the Box-and-Whisker Plot:**



The rectangle is drawn from the lower quartile (57) to the upper quartile (69) the vertical line in the middle is the median (61.5). The whiskers extend to the minimum value (40) and the maximum value (89).

OUTLIERS


Median
(Q_2)
Minimum Value in
the Data
25th Percentile (Q_1)
75th Percentile
(Q_3)
Maximum Value in
the Data
Potential
Outliers
Interquartile Range
(IQR)
Potential
Outliers
(Minimum Value in the Data, Q_1 – 1.5*IQR)
(Maximum Value in the Data, Q_3 + 1.5*IQR)

Example

- Draw a box plot for the following data:

55	48	42	75	40	49	53	50	52
50	5	46	58	56	102			

rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
data	5	40	42	46	48	49	50	50	52	53	55	56	58	75	102

Lower quartile

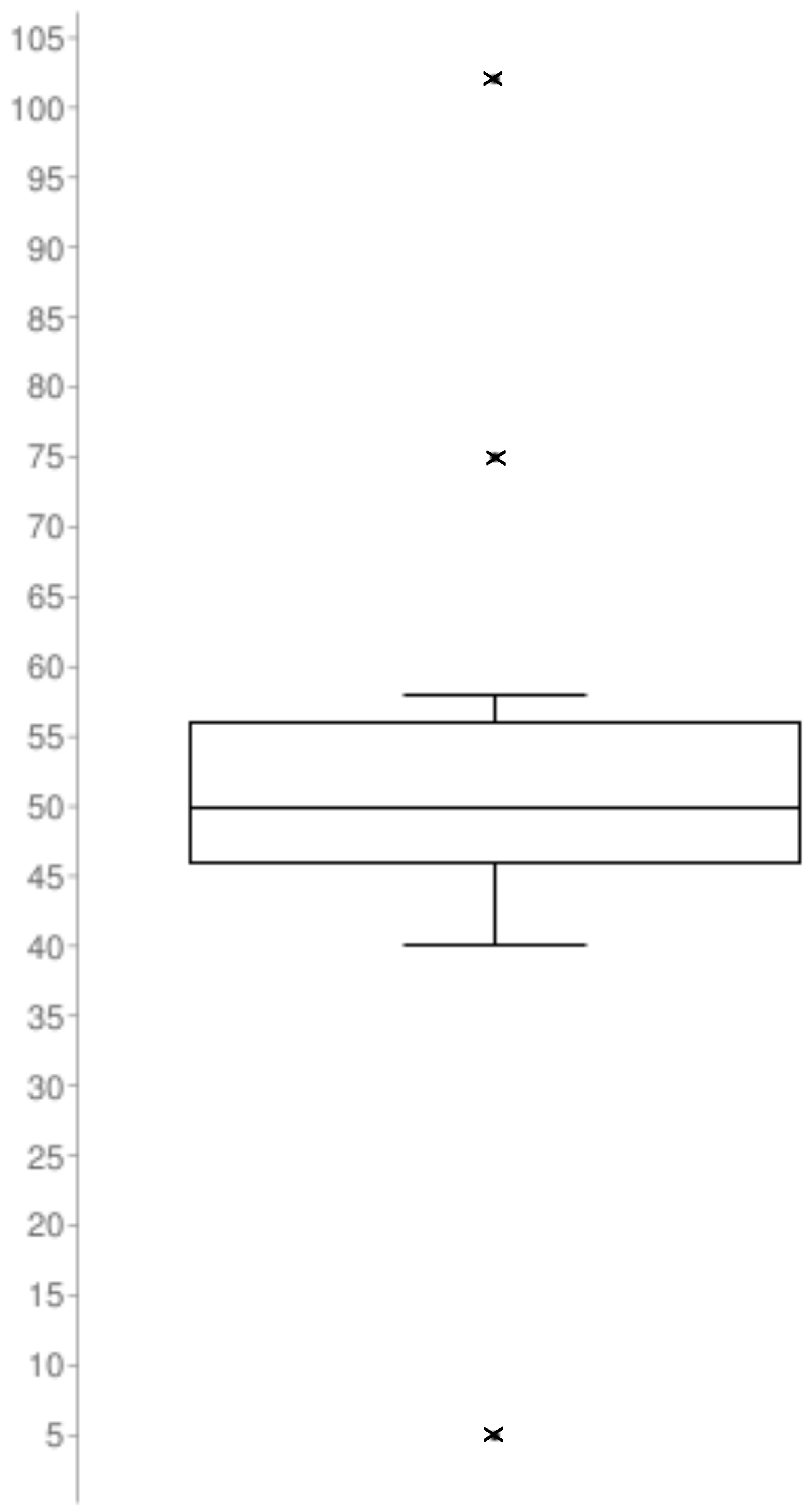
Median

Upper quartile

Inter quartile range (IQR) = 56 – 46 = 10

Maximum = 56 + 1.5 x IQR = 56 + 1.5 x 10 = 56 + 15 = 71

Minimum = 46 – 1.5 x IQR = 46 – 15 = 31


105
100
95
90
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5